# Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

## Static friction

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain static friction to a friend? | You push a crate gently and it stays still. What force balances your sideways push? |
| **Example or sketch** | **Non-example and why** |
| A stationary crate on a level floor is pushed right with 3 N below the sliding threshold. Draw friction. | The crate is already sliding across the floor. Would the friction at that contact still be static? |

## Limit

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain limit to a friend? | The floor can supply up to 6 N. How much friction balances a gentle 2 N horizontal push? |
| **Example or sketch** | **Non-example and why** |
| With a 6 N limit and no other horizontal forces, what do you predict for a 7 N push? | A friend writes 6 N for every trial, even with no sideways push. What does 6 N mean? |

# Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

## Kinetic friction

| | |
|---|---|
| **Meaning in my words**<br>How would you explain kinetic friction to a friend? | **How can you tell?**<br>A crate slides right across a stationary floor. Which way does kinetic friction act on the crate? |
| **Example or sketch**<br>If the coefficient is 0.2 and the normal force is 10 N, what arrow would you draw? | **Non-example and why**<br>A crate stays put while you push gently. Should you use the kinetic-friction rule? |

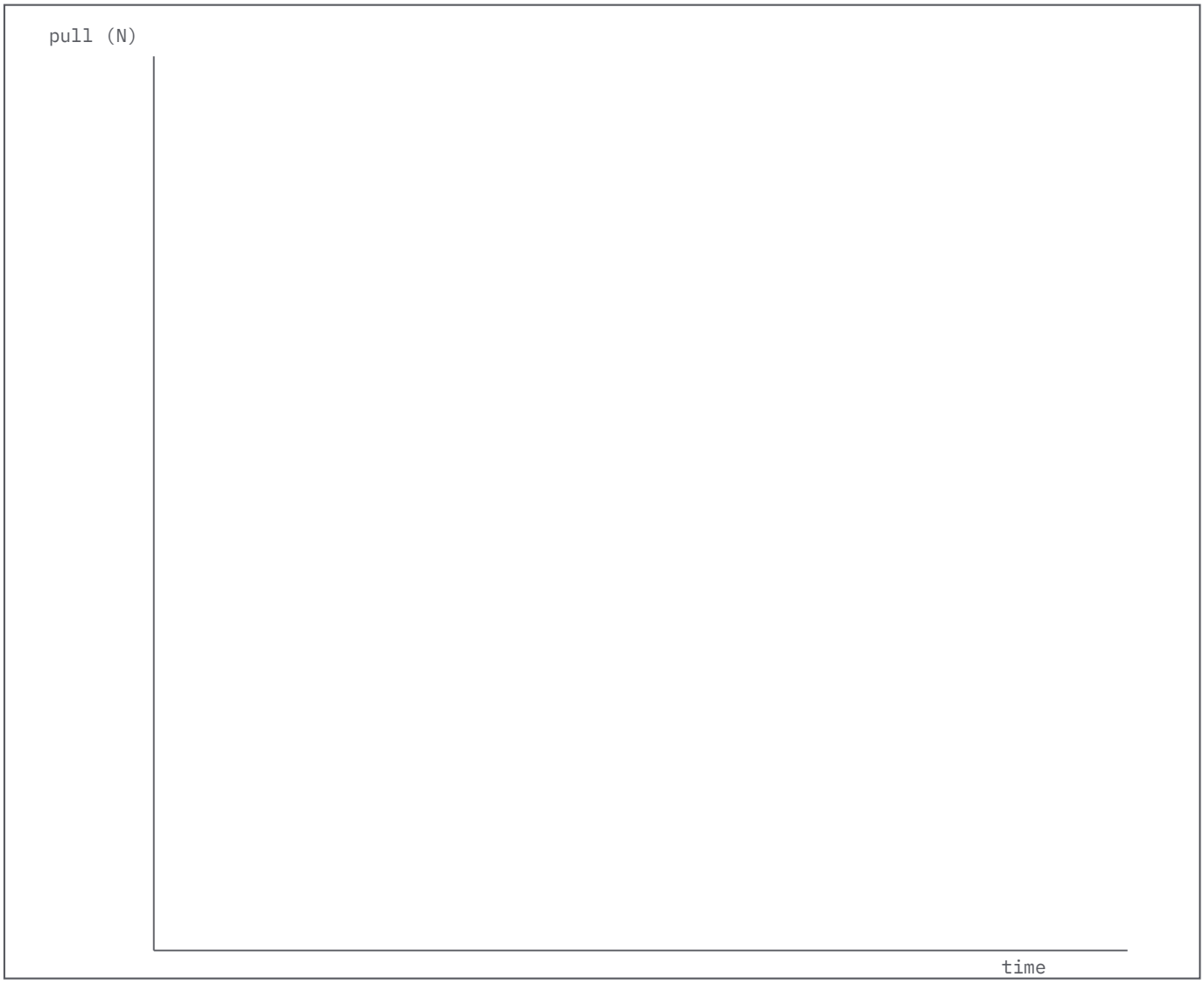
## Normal force

| | |
|---|---|
| **Meaning in my words**<br>How would you explain normal force to a friend? | **How can you tell?**<br>A crate rests on a ramp. Which way does the ramp push the crate? |
| **Example or sketch**<br>How would you draw the normal force on a book on a level table? | **Non-example and why**<br>Someone draws the normal force along the ramp. What would you rotate? |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Sketch a force trace. Label the peak just before slip and the steady-speed pull while the surfaces slide.




**My marks mean**

# Cut from the table to space

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**SECTION E** FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

**Our goal:** Use contact surfaces to decide when friction acts.

**CHOOSE ONE JOB TOGETHER**

**EFFECTS ENGINEER** ☐

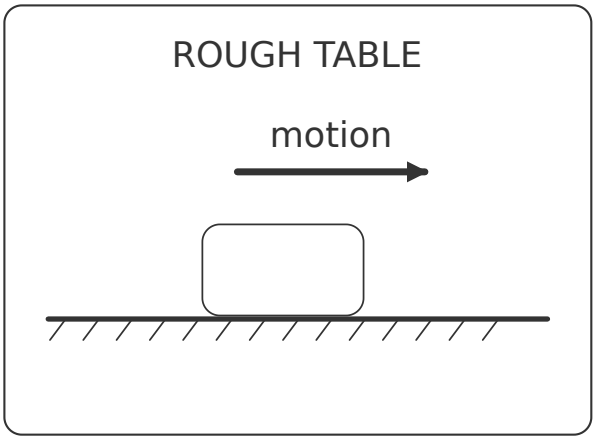
Set friction for each scene. Mark the touching surfaces and explain what could change XJ's motion in empty space.

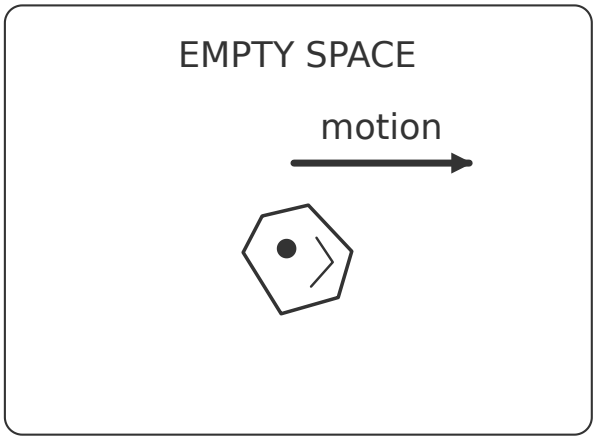
**SCIENCE BROADCASTER** ☐

Correct the caption for the two shots. Point to the surfaces and say what could still bend XJ's path.

Point to a clue. Keep your first marks; add the update using the color key.

A show cuts from a block sliding on a rough table to XJ coasting in empty space. The proposed caption says “Both will stop from surface friction.”




## 01 OUR FIRST CALL

**What should the two-scene caption say? Explain using the contact surfaces.**

## 02 OUR REVISED CALL

**Wait for the reveal. Keep your first call.**

New clue: _____

What changed in the new shot? Update the force explanation for XJ.

**Switch speakers.** How would entering an atmosphere change the forces you include? Explain where the added force comes from.

| ☐ GREEN | ☐ RED | ☐ YELLOW | ☐ BLUE | ☐ PURPLE | ☐ ORANGE | ☐ PENCIL |
|---|---|---|---|---|---|---|
| speeding up | slowing down | steady / balanced | position / displacement | velocity | forces | other marks |